

Abdullah Ali

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EDUCATION

Harvey Mudd College

Claremont, CA

Bachelor of Science in Computer Science and Mathematics

Relevant Coursework: CS 05 (Introduction to Computer Science), CS 60 (Principles of Computer Science, Algorithms & Complexity), Math 19 (Multivariable Calculus), Math 73 (Linear Algebra)

SKILLS

Languages: Python, Java, C++, Go, TypeScript, JavaScript, SQL, Racket, HTML, CSS

AI / ML: LLM Orchestration, Prompt Engineering, RAG Pipelines, Embeddings, Vector Databases (Qdrant), scikit-learn, NLP, Feature Engineering, Quantitative Signal Modeling

Backend & Cloud: FastAPI, Django, REST APIs, Docker, AWS (Fargate, S3), Terraform, CI/CD (GitHub Actions), Prometheus, Grafana, Redis, Celery, PostgreSQL, SQLite

Frontend & Tools: Vue.js, Next.js, Git, GitHub, Jupyter, NumPy, Pandas, CCXT

Quant & Systems: Time-Series Analysis, Backtesting, Walk-Forward Validation, Sharpe Ratio, Drawdown Analysis, Vectorization, Microservices, Data Structures & Algorithms

EXPERIENCE

Harvey Mudd College, Entrepreneurship Studio | AI Research Engineer | Claremont, CA

May 2025 – Present

- Architecting Lighthouse: a multi-tenant LLM visibility platform (FastAPI, Postgres + pgvector, arq workers) probing GPT-5, Claude, Perplexity Sonar, ChatGPT Search, and Gemini Grounding across versioned frozen query panels; RAG-grounded context ingestion (text-embedding-3-small, 1536-d HNSW index) lets marketing teams embed proprietary docs and measure causal content lift.
- Built a citation-classification and provenance-gap engine identifying which third-party sources drive AI recommendations and where the brand is absent; causal experiment layer uses difference-in-differences (targeted vs holdout queries) with z-test significance to isolate true lift from background engine drift.

Systems Limited | Cloud Application Development Intern | Lahore, Pakistan

December 2025 – January 2026

- Engineered scalable Python/Django backend services across 5+ production endpoints with optimized ORM data models for cloud-hosted enterprise applications.
- Automated CI/CD pipelines via Git to streamline testing and deployment across multi-environment systems, reducing release friction and improving delivery consistency.

Brainstorm | Software Engineering Intern | Claremont, CA

November 2025 – January 2026

- Architected responsive frontend interfaces using Vue.js and TypeScript with component-based state management, ensuring modularity and code reusability across the product.
- Evaluated 3+ UI architectures to identify access bottlenecks, improving content discoverability and reducing user friction across core workflows.

PROJECTS

Lighthouse | *Python, FastAPI, Postgres, pgvector, arq, Redis, Claude, GPT-5, Perplexity Sonar, Gemini*

- Multi-tenant GEO platform with RAG-grounded context ingestion (chunk + embed into pgvector HNSW), versioned frozen query panels, and async 5-engine probe fan-out; Haiku parser extracts brands, positions, attributes, and sentiment per response with citation-node classification and provenance-gap analysis.
- Crawler-readiness audits (httpx bot UAs + Playwright) surface AI-bot blockers per page; DiD causal experiment engine measures content lift on targeted vs holdout query sets with z-test significance; MCP server exposes tools to external agent.

QuantForge | *Python, scikit-learn, Pandas, NumPy, yfinance*

- Leakage-safe ML backtesting engine with past-only features, walk-forward validation, transaction-cost-aware simulation, and logistic/GBM models on price-based signals; yields Sharpe 1.10, +76% return, -17% max drawdown on momentum data, collapsing to a coin flip on a random walk, proving no lookahead leakage.

Anchor Infra | *Terraform, AWS ECS Fargate, ECR, ALB, GitHub Actions, Prometheus, Grafana*

- IaC provisioning a containerized RAG service on AWS (VPC across 2 AZs, private Fargate tasks, ALB, CPU autoscaling, ECR with image scanning, GitHub OIDC); CI/CD auto-deploys on push via OIDC short-lived tokens with Prometheus/Grafana and CloudWatch Container Insights for full observability.

ACTIVITIES & LEADERSHIP

Honors & Awards: International Youth Mathematics Challenge (IYMC) Gold Medalist | O Level Valedictorian (14 A/A*s) | Director's Award for Excellence in Information Technology (1/3600)

Leadership: Senior Prefect, International School Lahore (led conduct & safety for 700+ students; -30% repeat incidents) | Computer Science Society President (300+ members; ACM chapter; hackathons; PKR 1.2M raised) | ACM Chapter Founder & Mentor (ML/AI bootcamps; 100+ students mentored; 3x contest placements)

Campus Engagements: HMC Tour Guide, Upward Bound Program Tutor, Music Recital Usher

Interests: Poker & probability, strategy games, competitive programming, squash, tennis, badminton, hiking, books, travel